

İTÜ



Unsupervised Industrial Anomaly Detection

BLG561E - Deep Learning

Davide Colella - 912500295
Leonardo Russo - 912500322

10.06.2026

1. Motivation

2. Content

- CAE/AAE model
- Evaluation methods

3. Novelty

4. Results

- Image results
- Loss curves
- Evaluation Table

ist

al

—

+

Economic Impact of Automated Inspection

10-30%

Annual Revenue Drain

Cost of Poor Quality (COPQ) directly erodes profitability through scrap and rework. [1]

~80%

Human Accuracy Limit

Fatigue and subjectivity lead to critical defect escapes in manual inspection cycles.[2]

10x

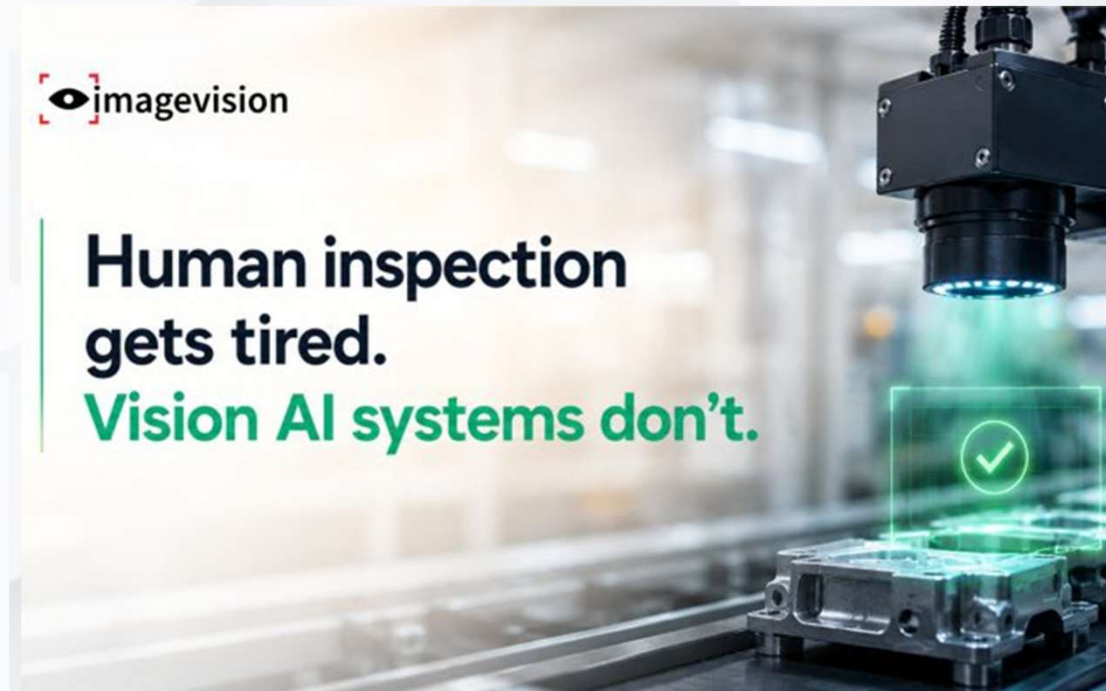
Failure Propagation

The cost to fix a defect increases tenfold at each subsequent stage of the value chain. [3]

+15%

OEE Improvement

AI integration reduces unplanned quality stops, maximizing Overall Equipment Effectiveness. [4]



report_problem

The Problem

Surface defects on production lines are:

- Rare and unpredictable
- Highly diverse in nature

Traditional supervised models fail due to the severe lack of labeled anomaly data.

lightbulb

Our Solution

Deep generative frameworks (CAE and AAE)

Trained exclusively on good (defect-free) samples.

auto_awesome

Key Highlights

- Adaptive threshold search space
- Connected-component spatial filtering

Eliminates background noise and false alarms.

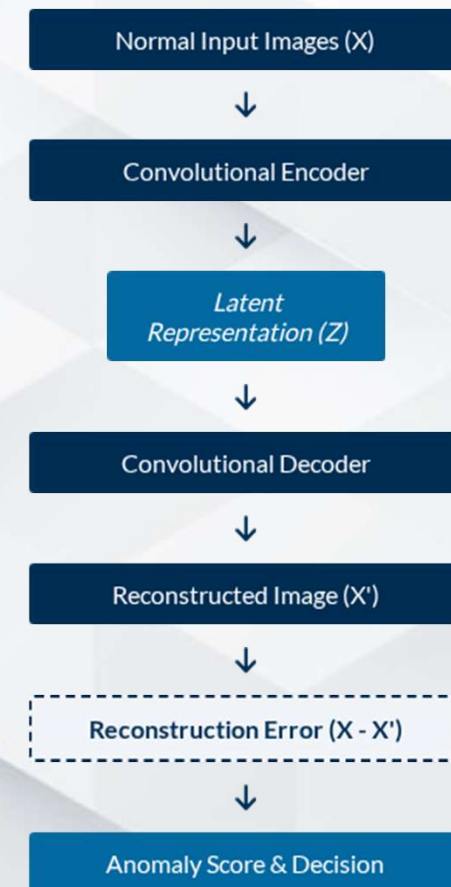


Content

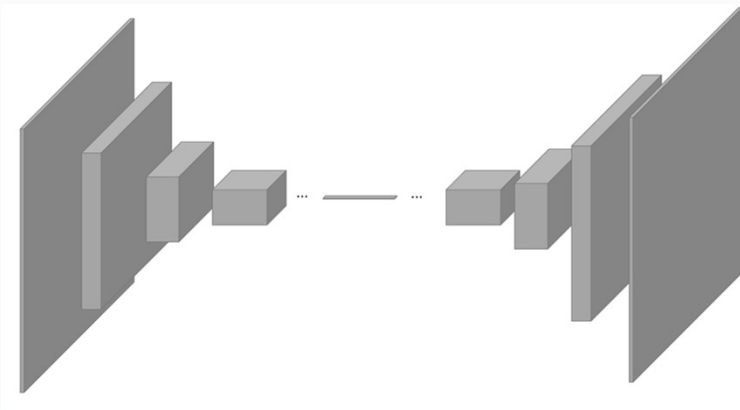
How it works

- ✓ **Unsupervised Advantage:** Models are trained only on "Normal" data, eliminating the need for rare defect labels.
- ✓ **Feature Bottleneck:** The latent representation forces the network to learn essential geometric and textural priors.
- ✓ **Reconstruction Gap:** Defects cannot be accurately reconstructed by a decoder that has only seen "Healthy" samples.
- ✓ **Scoring Mechanism:** L2 distance between input and output provides a pixel-wise heat-map of anomalies.

Technical Case



Convolutional Autoencoder (CAE)

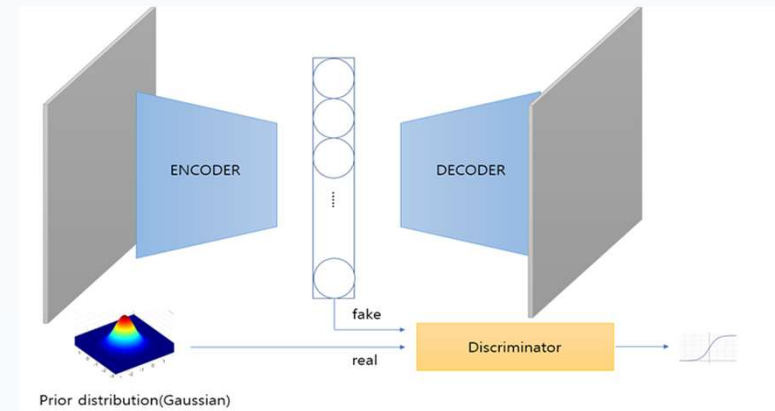


Paradigm: Deterministic reconstruction of input signals.

Objective: Minimizing reconstruction error (L2 or MS-SSIM).

Normalization: InstanceNorm2d (Encoder) / BatchNorm2d (Decoder).

Adversarial Autoencoder (AAE)



Paradigm: Generative with adversarial latent regularization.

Distribution: Latent space forced into a Multivariate Gaussian.

Errors: Encoder Loss, Discriminator Loss, Decoder Loss

Model	CAE	CAE	AAE	AAE
Loss function	L2	MSSIM	L2	MSSIM

Performance Evaluation

THRESHOLD-DEPENDENT

Pixel-Level Anomaly Score: Derived from reconstruction residuals using structural similarity.

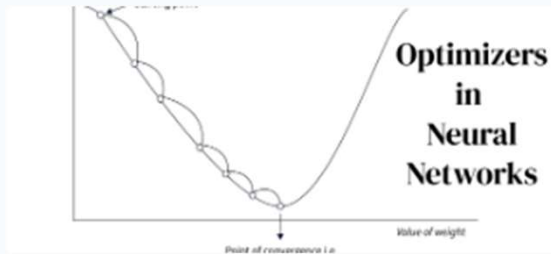
Binary Mask Generation: Applying critical thresholds (Otsu or fixed).

Spatial Filtering: Connected-component analysis to eliminate low-density noise artifacts.

THRESHOLD-INDEPENDENT

AU-ROC: Standard metric for overall classification performance between normal and anomalous sets.

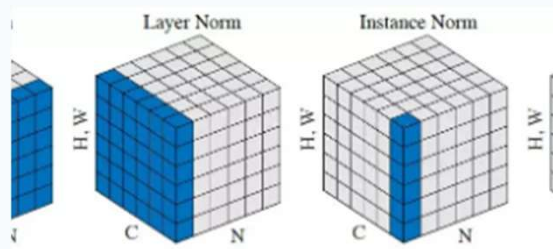
AU-PRO (Per-Region Overlap): The de facto standard for industrial defect detection (MVTec AD).



Adversarial Balance

Mitigating mode collapse in AAE training through structured constraints:

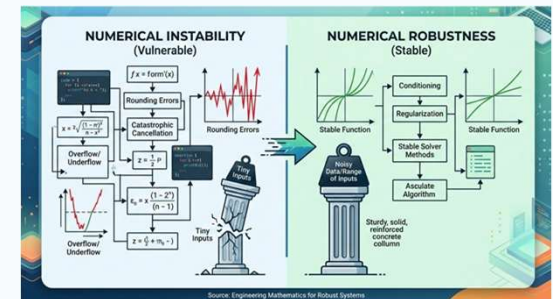
- **LR Asymmetry:** Discriminator learning rate is scaled by 1/5 ($\eta_{disc} = 0.2 * \eta_{base}$).
- **Gradient Clipping:** Max norm fixed at 1.0 to prevent gradient explosion.
- **Loss Smoothing:** Label smoothing (0.9 / 0.1) prevents absolute certainty in the discriminator.



Structural Asymmetry

Optimizing feature extraction and generation through specific layer selection:

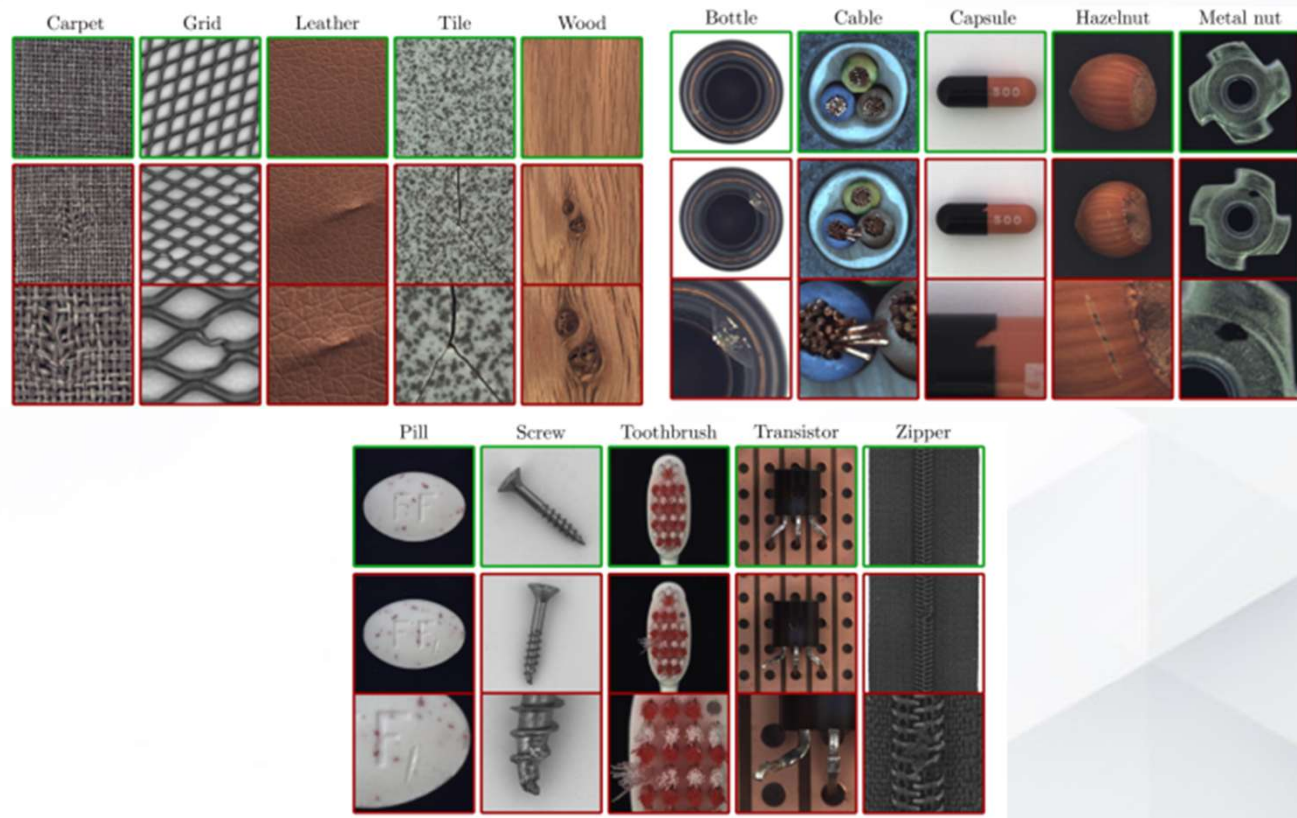
- **InstanceNorm (Encoder):** Eliminates style/lighting variations to focus solely on geometry.
- **BatchNorm (Decoder):** Stabilizes sample distribution during the generative pixel-reconstruction phase.
- **Weighted G-Loss:** Generator updates scaled to 0.05 for controlled latent optimization.



Numerical Robustness

Ensuring convergence and gradient health in the loss landscape:

- **MS-SSIM Anti-NaN Fix:** Implemented ReLU-based positive forcing for fractional power operations.
- **Multi-Scale Weighting:** Wang-schema normalized weights applied over 5 pooling levels.
- **Scaled L2:** MSE loss pre-multiplied by 10.0 to amplify small structural gradients.



MVTec AD Dataset

Category	Type	# Train (Good)	# Test (Good)	# Test (Defective / Bad)	Category Total
Carpet	Texture	280	28	89	397
Grid	Texture	264	21	57	342
Leather	Texture	245	32	92	369
Tile	Texture	230	33	84	347
Wood	Texture	247	19	60	326
Bottle	Object	209	20	63	292
Cable	Object	224	58	92	374
Capsule	Object	219	23	109	351
Hazelnut	Object	391	40	70	501
Metal Nut	Object	220	22	93	335
Pill	Object	267	26	141	434
Screw	Object	320	41	119	480
Toothbrush	Object	60	12	30	102
Transistor	Object	213	60	40	313
Zipper	Object	240	32	119	391
TOTAL		3,629	467	1,258	5,354



Novelty

Latent-Space Control (AAE vs. CAE)

- **The Context:** Standard Convolutional Autoencoders (CAEs) can sometimes yield reconstructions with slightly blurred pixels around complex edges, while pixel-level GANs can be highly unstable to train.
- **The Method:** To enhance structural sharpness, an Adversarial Autoencoder (AAE) model is implemented. A discriminator regularizes only the latent space, forcing it into a continuous distribution.
- **The Benefit:** The system smoothly accommodates minor normal variations (such as subtle lighting shifts) without edge blurring, substantially reducing false alarms

Automated Operational Thresholding

- **The Real-World Gap:** Academic benchmarks rely on theoretical, threshold-independent metrics (AUC-ROC). However, a factory line requires a single, concrete operational threshold to instantly accept or reject a part.
- **The Method:** A Binary Search calibration loop is executed entirely on the anomaly-free validation set. The algorithm dynamically tests trial thresholds and evaluates the resulting false-alarm blobs.
- **The Benefit:** The system automatically pinpoints the tightest mathematical boundary of "normality," balancing optimal defect sensitivity with zero structural false positives. Basically it solves the industrial trade-off by achieving **maximum defect sensitivity** with **zero false scraps**.

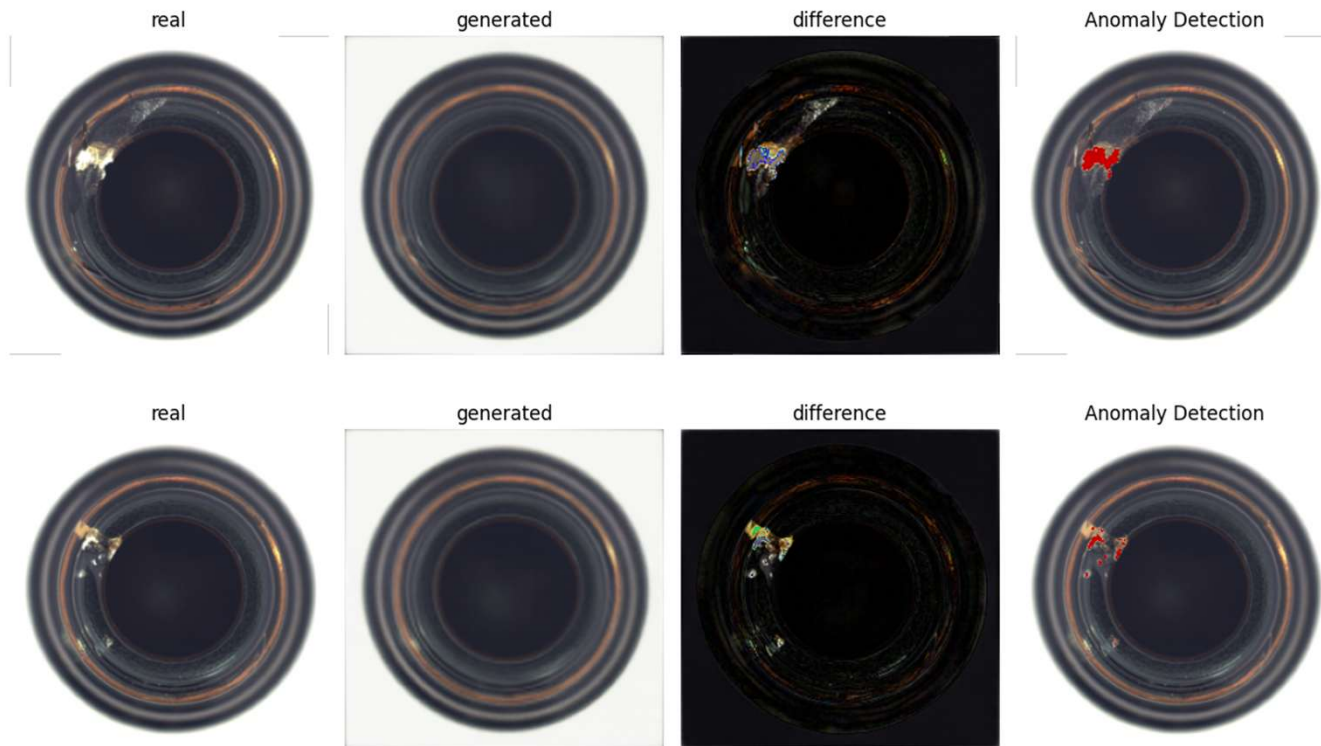
Adaptive Spatial Noise Filtering

- **The Problem:** Raw image subtraction generates pixel-level background noise (camera grain, micro-vibrations) that triggers frequent false positives.
- **The Method:** Adjacent error pixels are grouped into physical regions (connected components), enforcing a dynamic geometric constraint tailored to the resolution of each specific object category.
- **The Benefit:** False-alarm blobs smaller than the dynamically scaled minimum area (proportional to a 50-pixel baseline) are automatically ignored, ensuring only true structural defects are flagged.



Results/Conclusions

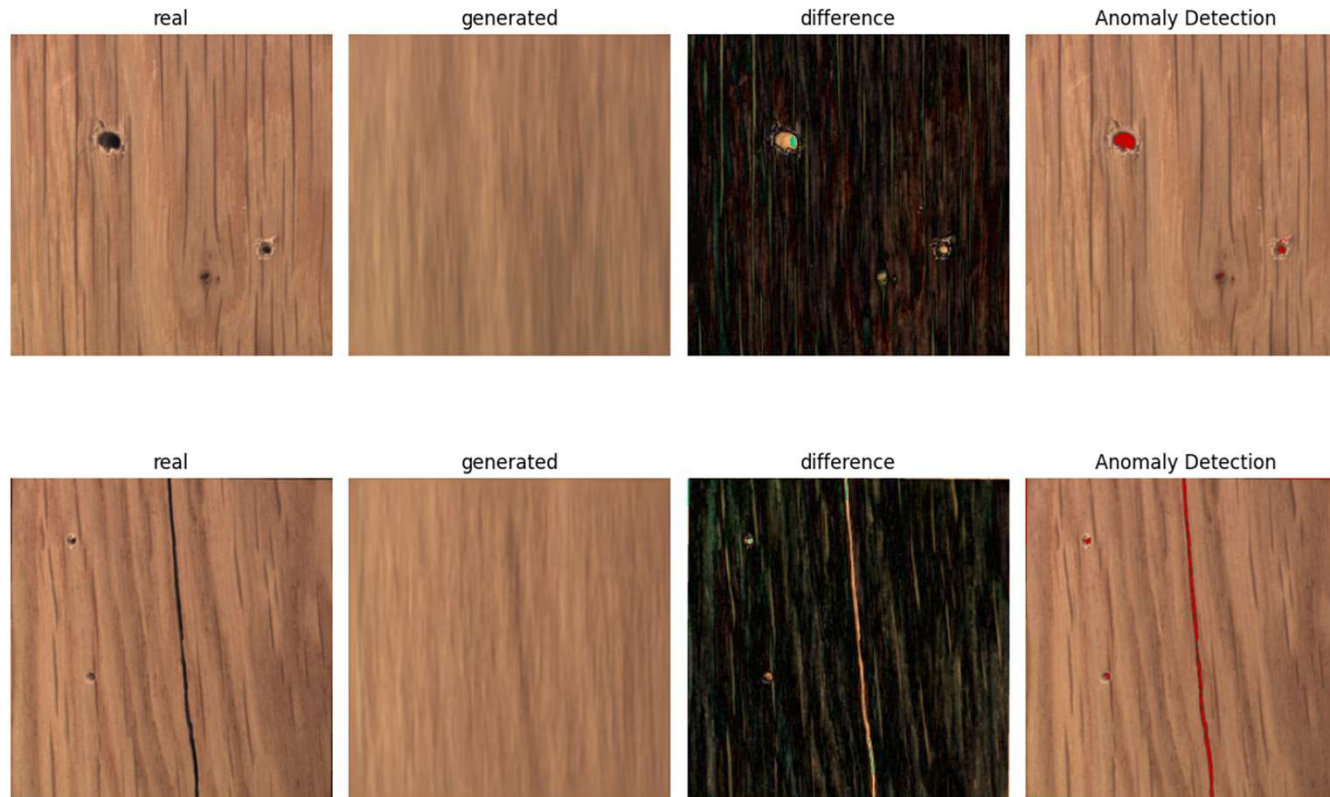
Results Bottle



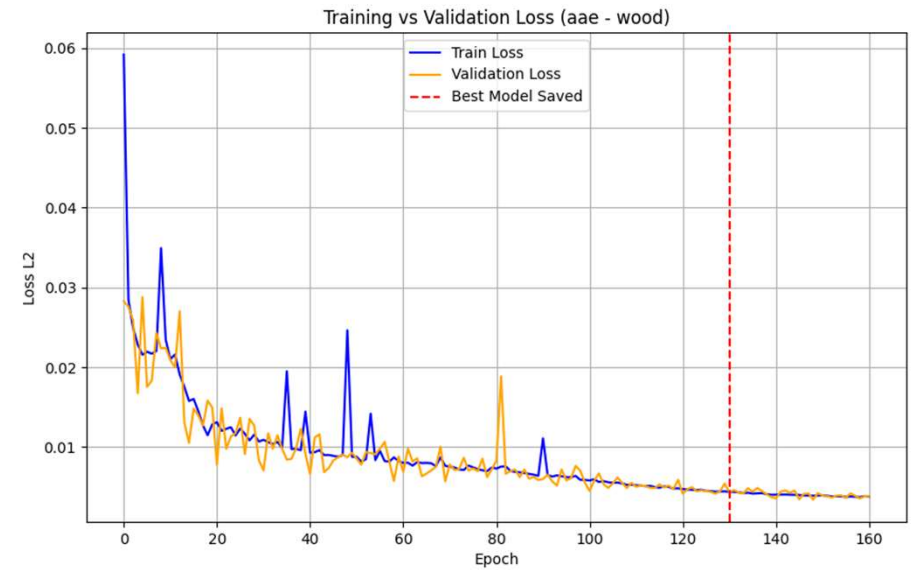
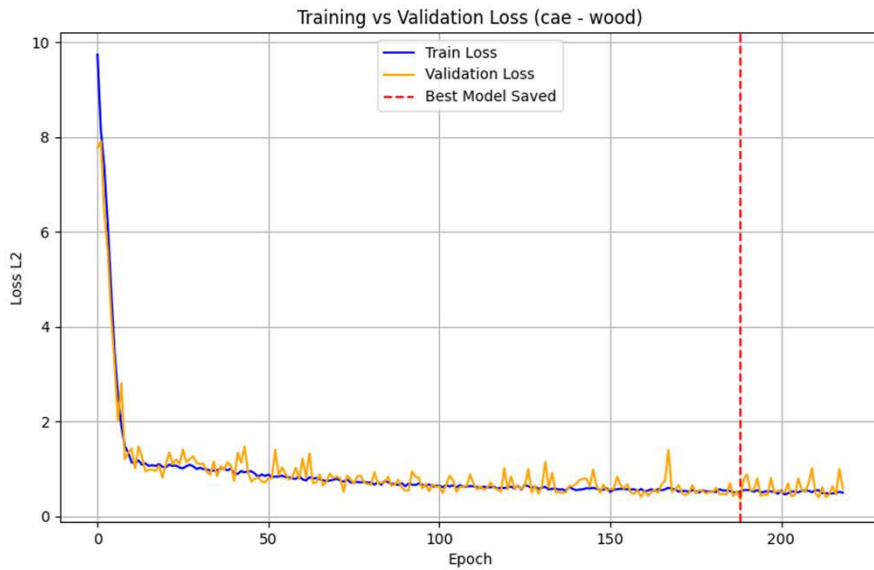
Results Capsule



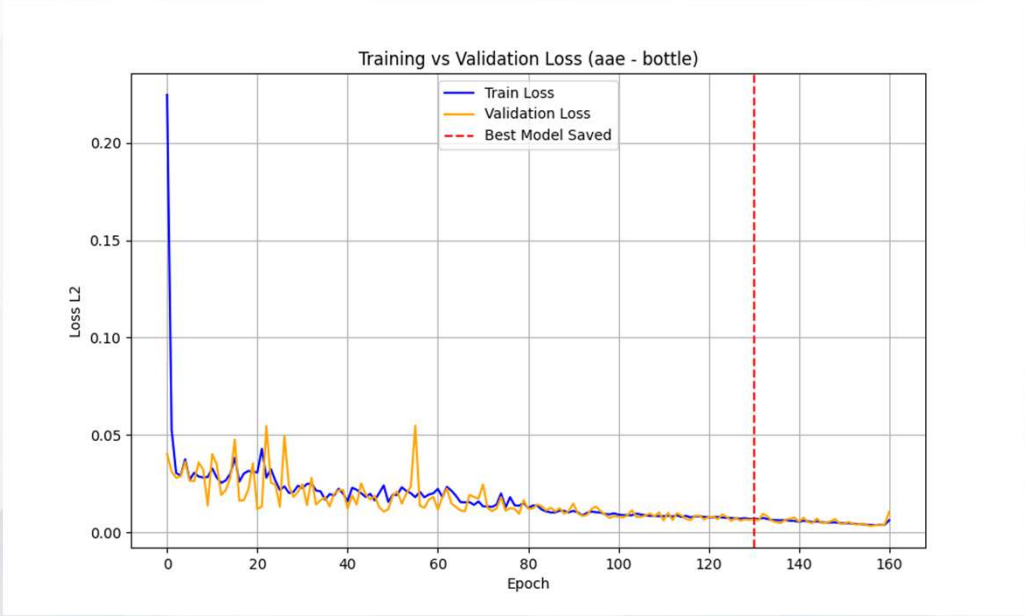
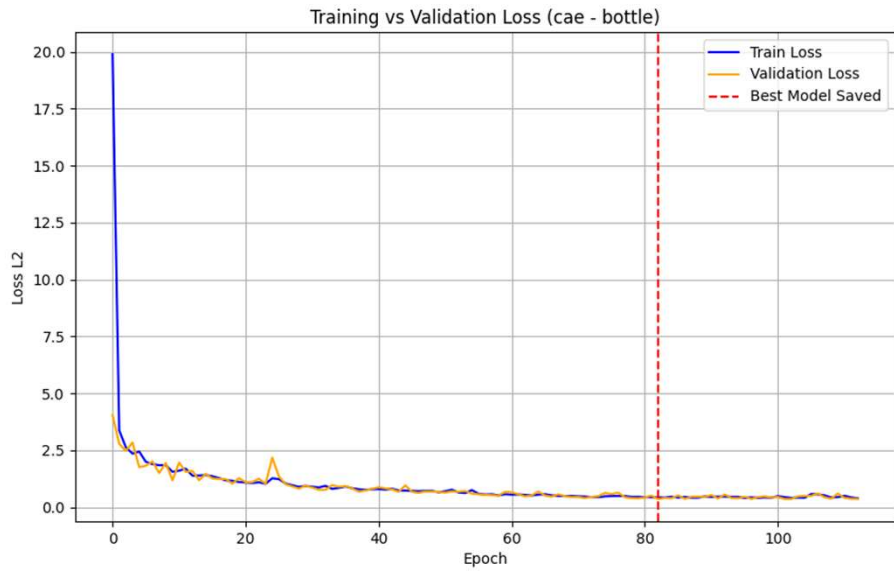
Results Wood



Loss Plot - Wood



Loss Plot - Bottle



Correctly classified samples of **anomaly-free** (top-row) and **anomalous** objects (bottom-row)

Object	AE (SSIM)	AE (L2)	AnoGAN	CAE (L2)	CAE (MSSIM)	AAE (L2)	AAE (MSSIM)
Bottle	0.85	0.7	0.95	0.00	0.45	0.0	0.55
	0.9	0.89	0.43	1.00	0.94	1.0	0.91
Capsule	0.78	1.00	0.96	0.61	0.22	0.04	0.52
	0.43	0.24	0.20	0.62	0.85	0.98	0.56
Wood	0.84	1.00	0.89	1.00	0.26	0.26	0.21
	0.82	0.47	0.47	0.73	0.93	1.00	0.95

Correctly classified samples of **anomaly-free** (top-row) and **anomalous** objects (bottom-row)

Object	AE (SSIM)	AE (L2)	AnoGAN	CAE (L2)	CAE (MSSIM)	AAE (L2)	AAE (MSSIM)
Bottle	0.85	0.7	0.95	0.00	0.45	0.0	0.55
	0.9	0.89	0.43	1.00	0.94	1.0	0.91
Capsule	0.78	1.00	0.96	0.61	0.22	0.04	0.52
	0.43	0.24	0.20	0.62	0.85	0.98	0.56
Wood	0.84	1.00	0.89	1.00	0.26	0.26	0.21
	0.82	0.47	0.47	0.73	0.93	1.00	0.95
Cable	0.74	0.93	0.98	0.0	0.0	0.01	0.0
	0.48	0.18	0.07	1.0	1.0	0.99	1.0

AU-PRO (top-row) and **AUC ROC** (bottom-row)

	MVTec			Our Method			
Object	AE (SSIM)	AE (L2)	AnoGAN	CAE (L2)	CAE (MSSIM)	AAE (L2)	AAE (MSSIM)
Bottle	0.15	0.22	0.05	0.59	0.57	0.56	0.60
	0.93	0.86	0.86	0.89	0.91	0.88	0.84
Capsule	0.09	0.11	0.04	0.47	0.48	0.11	0.35
	0.94	0.88	0.84	0.66	0.53	0.59	0.53
Wood	0.36	0.29	0.14	0.39	0.38	0.45	0.39
	0.73	0.73	0.62	0.95	0.70	0.88	0.56
Gr	0.01	0.05	0.01	0.52	0.50	0.47	0.42

- Implement other methods for threshold choice
- Use different different value of “minimum defect area”
- Train/Test on different objects/textures with different number of epochs
- Use other models (e.g. VAE)

İTÜ



Thank you for your attention!

BLG561E - Deep Learning

Davide Colella - 912500295
Leonardo Russo - 912500322

10.06.2026

ENCODER NETWORK SPECIFICATION

ENCODER

Layer Name	Layer	Input Shape	Output Shape	K	S	P	Filtri	Note / Attivazione
model.0	Conv2d	(3, 256, 256)	(16, 128, 128)	4×4	2	1	16	Estrazione iniziale
model.1	InstanceNorm2d	(16, 128, 128)	(16, 128, 128)	--	--	--	16	Affine = True
model.2	LeakyReLU	(16, 128, 128)	(16, 128, 128)	--	--	--	--	Slope = 0.2
model.3	Conv2d	(16, 128, 128)	(32, 64, 64)	4×4	2	1	32	--
model.4	InstanceNorm2d	(32, 64, 64)	(32, 64, 64)	--	--	--	32	Affine = True
model.5	LeakyReLU	(32, 64, 64)	(32, 64, 64)	--	--	--	--	Slope = 0.2
model.6	Conv2d	(32, 64, 64)	(64, 32, 32)	4×4	2	1	64	--
model.7	InstanceNorm2d	(64, 32, 32)	(64, 32, 32)	--	--	--	64	Affine = True
model.8	LeakyReLU	(64, 32, 32)	(64, 32, 32)	--	--	--	--	Slope = 0.2
model.9	Conv2d	(64, 32, 32)	(128, 16, 16)	4×4	2	1	128	--
model.10	InstanceNorm2d	(128, 16, 16)	(128, 16, 16)	--	--	--	128	Affine = True
model.11	LeakyReLU	(128, 16, 16)	(128, 16, 16)	--	--	--	--	Slope = 0.2
model.12	Conv2d	(128, 16, 16)	(256, 8, 8)	4×4	2	1	256	--
model.13	InstanceNorm2d	(256, 8, 8)	(256, 8, 8)	--	--	--	256	Affine = True
model.14	LeakyReLU	(256, 8, 8)	(256, 8, 8)	--	--	--	--	Slope = 0.2
model.15	Conv2d	(256, 8, 8)	(512, 4, 4)	4×4	2	1	512	--
model.16	InstanceNorm2d	(512, 4, 4)	(512, 4, 4)	--	--	--	512	Affine = True
model.17	LeakyReLU	(512, 4, 4)	(512, 4, 4)	--	--	--	--	Slope = 0.2
model.18	Conv2d	(512, 4, 4)	(1024, 1, 1)	4×4	1	0	1024	Riduzione a mappa unitaria

Nome Layer	Tipo Layer	Input Shape	Output Shape	K	S	P	Filtri	Attivazione / Note
model.0	ConvTranspose2d	(latent_dim, 1, 1)	(1024, 4, 4)	4×4	1	0	1024	LeakyReLU (True)
model.1	BatchNorm2d	(1024, 4, 4)	(1024, 4, 4)	--	--	--	1024	--
model.2	LeakyReLU	(1024, 4, 4)	(1024, 4, 4)	--	--	--	--	In-place
model.3	ConvTranspose2d	(1024, 4, 4)	(512, 8, 8)	4×4	2	1	512	LeakyReLU (True)
model.4	BatchNorm2d	(512, 8, 8)	(512, 8, 8)	--	--	--	512	--
model.5	LeakyReLU	(512, 8, 8)	(512, 8, 8)	--	--	--	--	In-place
model.6	ConvTranspose2d	(512, 8, 8)	(256, 16, 16)	4×4	2	1	256	LeakyReLU (True)
model.7	BatchNorm2d	(256, 16, 16)	(256, 16, 16)	--	--	--	256	--
model.8	LeakyReLU	(256, 16, 16)	(256, 16, 16)	--	--	--	--	In-place
model.9	ConvTranspose2d	(256, 16, 16)	(128, 32, 32)	4×4	2	1	128	LeakyReLU (True)
model.10	BatchNorm2d	(128, 32, 32)	(128, 32, 32)	--	--	--	128	--
model.11	LeakyReLU	(128, 32, 32)	(128, 32, 32)	--	--	--	--	In-place
model.12	ConvTranspose2d	(128, 32, 32)	(64, 64, 64)	4×4	2	1	64	LeakyReLU (True)
model.13	BatchNorm2d	(64, 64, 64)	(64, 64, 64)	--	--	--	64	--
model.14	LeakyReLU	(64, 64, 64)	(64, 64, 64)	--	--	--	--	In-place
model.15	ConvTranspose2d	(64, 64, 64)	(32, 128, 128)	4×4	2	1	32	LeakyReLU (True)
model.16	BatchNorm2d	(32, 128, 128)	(32, 128, 128)	--	--	--	32	--
model.17	LeakyReLU	(32, 128, 128)	(32, 128, 128)	--	--	--	--	In-place
model.18	ConvTranspose2d	(32, 128, 128)	(3, 256, 256)	4×4	2	1	3	LeakyReLU (True)
model.19	Tanh	(3, 256, 256)	(3, 256, 256)	--	--	--	--	Output limitato in [-1, 1]